EXPERIMENT NO 4

```
#include <iostream>
#include <vector>
using namespace std;
int knapsack(int W, vector<int> &wt, vector<int> &val, int n) {
   // Create DP table
   vector<vector<int>> dp(n + 1, vector<int>(W + 1, 0));
   // Build table dp[][] in bottom-up manner
   for (int i = 0; i <= n; i++) {
      for (int w = 0; w <= W; w++) {
         if (i == 0 || w == 0)
            dp[i][w] = 0;  // Base case
         else if (wt[i - 1] <= w)
            dp[i][w] = max(val[i - 1] + dp[i - 1][w - wt[i - 1]], dp[i - 1][w]);
         else
            dp[i][w] = dp[i - 1][w];
      }
   }
   // dp[n][W] contains the maximum value
   return dp[n][W];
}
int main() {
   int n, W;
   cout << "Enter number of items: ";
   cin >> n;
   vector<int> val(n), wt(n);
   cout << "Enter value and weight of each item:\n";
   for (int i = 0; i < n; i++)
      cin >> val[i] >> wt[i];
```

```
    cout << "Enter capacity of knapsack: ";
    cin >> W;
    int maxValue = knapsack(W, wt, val, n);
    cout << "Maximum value in 0-1 Knapsack = " << maxValue << endl;
    return 0;
}
```

OUTPUT

```
icoer@icoer-Vostro-3470:~$ g++ 0-1_knapsack.cpp
icoer@icoer-Vostro-3470:~$ ./a.out
Enter number of items: 3
Enter value and weight of each item:
100 10
120
30
20
130
Enter capacity of knapsack: 50
Maximum value in 0-1 Knapsack = 220
```